

08.10.2026 items review

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1 Status of the 02.10 items

1. **Cluster and emulator. Done.** Libraries installed.
2. **Weight precision study ($\pm 5\%$). Open.**
3. **Interference cost β_{nm} . Done.** Each antenna has its own band, so there is no interference between antennas. Inside an antenna, adjacent channel leakage is a soft edge J_{uv} (cost if both): the spectral efficiency the pair loses by sharing the antenna.
4. **Embedding. Open.**
5. **Post processing proof. Reframed.** Post processing is now two steps: a prune to capacity C and a coverage repair.
6. **Realistic problem. Done.** Each antenna has a 20MHz band split into $C = 6$ channels, so it serves up to six users per slot. Constraints: blockage (hard) and leakage (soft). The scenario is dynamic: 20-50% of users arrive or move between slots. Pilot contamination is no longer modelled: we assume enough pilots.
7. **Reformulation for $N > 1$. Done.** Nodes are UEs, as proposed, but instead of colours = $(AP, \text{timeslot})$ there is one small graph per antenna and timeslot (n^a antennas x n^o timeslots). The handover cost is a continuity bonus in the node weight.

2 Context

1. **Why one graph per antenna.** The $(UE, \text{timeslot})$ graph allows $N=1$ or $N=\infty$, not $N > 1$. It also grows with $K \cdot T$ (1,200 nodes at $K = 120, T = 10$). Splitting by antenna and timeslot keeps every problem within the emulator limit of about 20-25.

2. **One antenna as a graph.** $x_u = 1$ if the antenna serves user u :

$$\max \sum_u w_u x_u - \sum_{(u,v) \text{ soft}} J_{uv} x_u x_v,$$

where

- $w_u = \text{SE}(u, a)/5.55 + 0.5 \cdot c_u$, with $c_u = 1$ if antenna a served u in the previous slot (continuity);
- $J_{uv} = 2 \cdot (\text{SE the pair loses to leakage})/5.55$;
- hard edges join users closer than 25 m.

3. **Capacity is left out of the iterations.** A classical prune keeps at most C users.

4. **Coverage by post processing.** Antennas are processed in order, users left out are served by a min cost max flow that moves as few users as possible. It was needed in majority of slots.

5. **Greedy results** ($K = 120$, $L = 30$, $C = 6$, $T = 10$, seed 2). Valid in every slot, median 15.6 Mbps and worst 10% at 9.6 Mbps per user. Jain index 0.942, 19% handover, and all 175 new users were served in their first slot.

3 Items

1. Tune the weights of quality, continuity and leakage. (might leave for the end if there's time so I don't focus too much on the telecommunications problem instead of the quantum)
2. Decide whether capacity goes in the QUBO. (Wesley recommended to keep it outside)
3. Build and evaluate the QUBO.
4. Try the QUBO solver on simpler, smaller problems to learn how it works.
5. Carried over: weight precision study ($\pm 5\%$ perturbations) on the new QUBOs. (Not imminent)
6. Carried over: show if the prune and the repair affect solution quality.

4 Benchmarking plan

Same scenario and random seed used on every solver; every schedule is checked with the same validity rules.

- Greedy oracle: the baseline.
- QUBO/MIS, solved classically (checks the QUBO formulation agrees with Greedy).
- Pulser based quantum sampling on the emulator/QPU.

Metrics: gap to the exact optimum per QUBO; total spectral efficiency, coverage (must be 100%), worst, 10th percentile and median user rate, Jain index, handovers and new users served, runtime against QUBO size.